

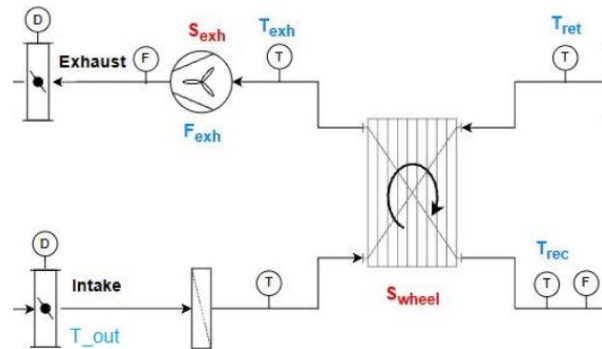
1. Problem State

Two models

- Heat exchanger: predicts post-recovery temperature $T_{rec}(t+1)$

Inputs: T_{out} , T_{ret} , T_{rec} , pressure_hex

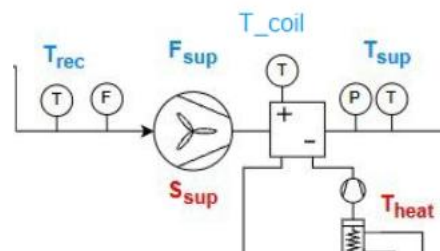
Outputs: $T_{rec}(t+1)$



- Heating coil: predicts supply air temperature $T_{sup}(t+1)$

Inputs: T_{rec} , T_{coil} , T_{sup}

Outputs: $T_{sup}(t+1)$



No actuator or control signals are used

All predictions are one-step-ahead (5-minute horizon).

	timestamp	T_{out}	T_{ret}	T_{rec}	T_{coil}	T_{sup}	pressure_hex	u_heat	u_heat_recovery	u_FF1	is_running
1	2021-11-26 05:00:00	2.54286	22.1	16.2857	32.1615	32.0846	21.45	100	100	47	1
2	2021-11-26 07:05:00	1.82333	22.1571	14.9	26.825	30.9	51.5	100	100	60	1
3	2021-11-26 07:10:00	1.86667	22.1557	14.9	26.8667	30.9	51.5	100	100	60	1
4	2021-11-26 07:15:00	1.89815	22.1543	14.9333	26.6667	30.7167	51.5	100	100	60	1
5	2021-11-26 07:20:00	1.92963	22.1529	14.9	26.7571	30.7625	51.5	100	100	60	1
6	2021-11-26 07:25:00	1.96111	22.1514	14.9352	26.4	30.4667	51.5	100	100	60	1
7	2021-11-26 07:30:00	1.99259	22.15	14.9704	26.6	30.525	51.5333	100	100	60	1
8	2021-11-26 07:35:00	2.02407	22.1429	15.0056	26.5	30.4	51.5	100	100	60	1
9	2021-11-26 07:40:00	2.05556	22.15	15.0407	26.1667	30.1333	51.5	100	100	60	1
10	2021-11-26 07:45:00	2.08704	22.1556	15.0759	26.45	30.14	51.5	100	100	61	1
11	2021-11-26 08:55:00	2.8	22.15	15.3727	26.0833	29.7	51.5556	100	100	60	1
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Prediction strategy: rather than predicting absolute values directly, all models predict the delta $\Delta = X(t+1) - X(t)$, then reconstruct the absolute value as:

$$X_{pred}(t+1) = X(t) + \Delta_{pred}$$

Baseline (persistence): $X_{pred}(t+1) = X(t)$, i.e. $\Delta_{pred} = 0$.

Primary Evaluation Metric "Closer-to" Analysis

For each predicted value $R(t+1)$:

Is it closer to the true next value $X(t+1)$, or closer to the previous value $X(t)$?

$\text{closer_to_}X(t+1): |R(t+1) - X(t+1)| < |R(t+1) - X(t)| \rightarrow \text{model beats persistence}$

$\text{closer_to_}X(t): |R(t+1) - X(t+1)| > |R(t+1) - X(t)| \rightarrow \text{model no better than persistence}$

2. Method Versions

V1 — ARX Baseline with Lag Features

Approach:

- Models: Ridge regression (ARX) + residual tree ensemble (18 trees, max_depth=8)
- Uniform sample weighting (standard MSE minimisation)

Results on validation set (5-min horizon):

T_rec	ridge_arx:	$\text{closer_to_}X(t+1) = 29.2\%$	$\text{closer_to_}X(t) = 55.8\%$
T_rec	tree_ensemble:	$\text{closer_to_}X(t+1) = 27.9\%$	$\text{closer_to_}X(t) = 57.1\%$
T_sup	ridge_arx:	$\text{closer_to_}X(t+1) = 21.6\%$	$\text{closer_to_}X(t) = 74.4\%$
T_sup	tree_ensemble:	$\text{closer_to_}X(t+1) = 14.6\%$	$\text{closer_to_}X(t) = 81.3\%$

T_rec validation MAE: ridge = 0.0989, tree = 0.0978 (persistence = 0.1049)

T_sup validation MAE: ridge = 0.1243, tree = 0.1180 (persistence = 0.1294)

Shortcomings:

1. Only 3 lag steps (15 min) — insufficient to capture trend direction

V2 — Sequence Window + Subsystem-Specific Inputs

Approach:

- Extended history window from 3 lag rows to $[t-12 \dots t]$ (13 steps = 1 hour)
- Separated subsystem inputs:
 - heat_exchanger: $T_{\text{out}}, T_{\text{ret}}, T_{\text{rec}}, \text{pressure_hex}$ (104 features)
 - heating_coil: $T_{\text{rec}}, T_{\text{coil}}, T_{\text{sup}}$ (79 features)
- Models: weighted Ridge + tree ensemble (same as v1)

Results on validation set (5-min horizon):

T_rec	seq_linear:	$\text{closer_to_}X(t+1) = 27.8\%$	$\text{closer_to_}X(t) = 57.2\%$
T_rec	seq_tree:	$\text{closer_to_}X(t+1) = 24.7\%$	$\text{closer_to_}X(t) = 60.3\%$
T_sup	seq_linear:	$\text{closer_to_}X(t+1) = 21.3\%$	$\text{closer_to_}X(t) = 74.7\%$
T_sup	seq_tree:	$\text{closer_to_}X(t+1) = 11.6\%$	$\text{closer_to_}X(t) = 84.4\%$

T_rec validation MAE: linear = 0.0982, tree = 0.1007 (persistence = 0.1048)

T_sup validation MAE: linear = 0.1271, tree = 0.1251 (persistence = 0.1340)

Shortcomings:

1. The closer-to metric did not improve compared to v1, and in some cases worsened.
Extending the history window alone does not solve the root problem.
2. Root cause identified: class imbalance in the delta distribution.
Approximately 54% of T_rec training deltas have $|\text{delta}| < 0.05 \text{ degC}$.
Under standard MSE, the model learns that predicting near-zero delta (persistence) minimises the loss — regardless of the input features.

V3 — Linear Sample Weighting (addressing class imbalance)

Improving Approach:

- Sample weights applied during training: $w = 1 + k * |\text{delta}|$
Samples with larger delta receive proportionally more weight, forcing the model to pay attention to transitional periods rather than just stable periods.
- Three values of k tested: k = 5, 10, 20
- Weighted Ridge: solved via $X^T W X$ formulation (exact weighted least squares)
- Weighted tree ensemble: weighted bootstrap sampling + weighted SSE node splitting

Results on validation set (5-min horizon), closer_to_X(t+1):

k = 5:

T_rec w_ridge: 35.3% w_tree: 35.2%

T_sup w_ridge: 33.3% w_tree: 28.3%

k = 10:

T_rec w_ridge: 37.1% w_tree: 37.7%

T_sup w_ridge: 34.6% w_tree: 34.2%

k = 20:

T_rec w_ridge: 38.9% w_tree: 40.6%

T_sup w_ridge: 36.1% w_tree: 33.7%

T_rec validation MAE (k=20): w_ridge = 0.1088, w_tree = 0.1244 (persistence = 0.0916)

T_sup validation MAE (k=20): w_ridge = 0.2099, w_tree = 0.2775 (persistence = 0.1471)、

Outcomes:

The closer-to metric improved substantially with increasing k, confirming that sample weighting is an effective strategy for this class imbalance problem.

Shortcomings:

1. Linear weight growth means very large deltas receive extremely high weights (e.g. at k=20, max weight exceeds 100x the minimum), potentially over-amplifying noisy outlier samples.
2. The weight function does not distinguish between "near-zero, safely ignorable" deltas and "moderate" deltas — there is no floor for suppressing stable samples explicitly.
3. T_sup remains harder to improve — the heating coil response is less predictable from sensor data alone.

V4 — Piecewise Sample Weighting

Improving Approach:

- Replaced linear weight with a two-segment piecewise function:

if $|\delta| < \text{threshold}$: $w = w_{\text{min}}$

else: $w = 1 + k * |\delta|^{\text{power}}$

Default parameters: threshold = 0.05, $w_{\text{min}} = 0.1$, $k = 3.0$, power = 2.0

Interpretation:

- Near-stable steps ($|\delta| < 0.05$ degC, ~54% of T_{rec} samples) are heavily discounted to weight 0.1, preventing the model from over-learning persistence.
- Transitional steps receive a quadratic boost, which is gentler than linear for mid-range deltas but still amplifies large transitions.

Weight values at selected delta magnitudes:

$|\delta| = 0.01$: $w = 0.10$ (suppressed)

$|\delta| = 0.05$: $w = 1.01$ (boundary)

$|\delta| = 0.20$: $w = 1.12$

$|\delta| = 0.50$: $w = 1.75$

$|\delta| = 1.00$: $w = 4.00$

Results on validation set (5-min horizon):

T_{rec}	pw_ridge:	closer_to_X(t+1) = 41.7%	closer_to_X(t) = 45.5%	tied = 12.8%
T_{rec}	pw_tree:	closer_to_X(t+1) = 37.1%	closer_to_X(t) = 50.1%	tied = 12.8%
T_{sup}	pw_ridge:	closer_to_X(t+1) = 38.5%	closer_to_X(t) = 56.4%	tied = 5.1%
T_{sup}	pw_tree:	closer_to_X(t+1) = 29.5%	closer_to_X(t) = 65.5%	tied = 5.1%

Results on training set (5-min horizon):

T_{rec}	pw_ridge:	closer_to_X(t+1) = 46.1%	closer_to_X(t) = 41.7%	(FLIPPED)
T_{rec}	pw_tree:	closer_to_X(t+1) = 41.6%	closer_to_X(t) = 46.2%	
T_{sup}	pw_ridge:	closer_to_X(t+1) = 37.7%	closer_to_X(t) = 56.8%	
T_{sup}	pw_tree:	closer_to_X(t+1) = 29.3%	closer_to_X(t) = 65.3%	

T_{rec} validation MAE: pw_ridge = 0.1332, pw_tree = 0.1183 (persistence = 0.0916)

T_{sup} validation MAE: pw_ridge = 0.2932, pw_tree = 0.2466 (persistence = 0.1471)

Notable observations:

- pw_ridge achieves the best closer_to_X(t+1) on validation for T_{rec} (41.7%), surpassing v3 k=20 (38.9%), meaning more predictions close to the true value.
- On the training set, pw_ridge T_{rec} has FLIPPED: 46.1% > 41.7%, meaning the model genuinely learns to predict the next state better than persistence on seen data.
- The gap between train (46.1%) and validation (41.7%) indicates mild generalisation loss — the model learns large-delta patterns well on training data but these do not fully transfer to unseen periods.

- pw_tree underperforms pw_ridge across all conditions, suggesting Ridge regression responds more effectively to the piecewise weighting scheme for this dataset.

Shortcomings:

1. On validation, closer_to_X(t+1) still does not exceed 50% for either target — the model has not fully overcome the persistence tendency.
2. MAE/RMSE are considerably higher than persistence, reflecting the accuracy-responsiveness trade-off inherent to aggressive sample weighting.
3. T_sup prediction remains significantly harder: even the best model (pw_ridge, 38.5%) still has 56.4% of predictions closer to X(t), meaning the heating coil subsystem is not yet reliably modelled without control signals.

5. KEY FINDINGS AND LIMITATIONS

1. Delta prediction + sample weighting is the most impactful design decision.
2. The piecewise weight function outperforms linear weighting.
3. Ridge regression outperforms tree ensemble under aggressive weighting.
4. T_sup is harder to model than T_rec without control signals.
5. No validation-set closer_to_X(t+1) ratio exceeds 50%.